

Missing Value Check - Raw Data

	<i>N</i>	
<i>Variable</i>	<i>Miss</i>	<i>N</i>
litter	0	79
trt	0	79
wgtgain	4	75

Range Check - Weight Gain Variable (Raw)

Analysis Variable : wgtgain				
Minimum	Maximum	Mean	Std Dev	N
-3.2000000	99.5000000	21.6069333	9.7298128	75

Frequency Check - Treatment Groups and Litter IDs (Raw)

<i>trt</i>	<i>Frequency</i>	<i>Percent</i>
1	9	11.39
2	9	11.39
3	10	12.66
4	8	10.13
5	8	10.13
6	8	10.13
7	9	11.39
8	9	11.39
9	8	10.13
10	1	1.27

<i>litter</i>	<i>Frequency</i>	<i>Percent</i>
1	9	11.39
2	10	12.66
3	10	12.66
4	10	12.66
5	11	13.92
6	10	12.66
7	9	11.39
8	9	11.39
9	1	1.27

Duplicate Records Found

<i>Obs</i>	<i>litter</i>	<i>trt</i>	<i>wgtgain</i>
1	2	1	19.86
2	4	7	99.5
3	5	3	23.96
4	5	8	.
5	6	3	-3.2

Verification - Cleaned Dataset

Variable	N		Minimum	Maximum	Mean	Std Dev
	Miss	N				
litter	0	69	1.0000000	8.0000000	4.5072464	2.3491124
trt	0	69	1.0000000	9.0000000	4.9855072	2.6205789
wgtgain	0	69	16.4400000	26.3800000	20.9268116	1.9526727

Frequency Check - Treatment Groups and Litter IDs (Cleaned)

<i>trt</i>	<i>Frequency</i>	<i>Percent</i>
1	8	11.59
2	8	11.59
3	7	10.14
4	8	11.59
5	7	10.14
6	8	11.59
7	8	11.59
8	7	10.14
9	8	11.59

<i>litter</i>	<i>Frequency</i>	<i>Percent</i>
1	9	13.04
2	9	13.04
3	8	11.59
4	9	13.04
5	7	10.14
6	9	13.04
7	9	13.04
8	9	13.04

Frequency Check - Treatment Groups and Litter IDs (Cleaned)

<i>Analysis Variable : wgtgain</i>						
<i>trt</i>	<i>N</i> <i>Obs</i>	<i>Mean</i>	<i>Std Dev</i>	<i>Minimum</i>	<i>Maximum</i>	<i>N</i>
1	8	22.4175000	2.4471076	18.7500000	26.3800000	8
2	8	21.5425000	1.7331289	19.1300000	23.9500000	8
3	7	20.3428571	2.3022287	17.3000000	23.9600000	7
4	8	19.9637500	0.7482539	19.0200000	21.1600000	8
5	7	20.7771429	2.2874419	16.4400000	23.7100000	7
6	8	21.9462500	1.9658218	18.6800000	25.0400000	8
7	8	19.9650000	1.5524819	17.3800000	21.9800000	8
8	7	19.8900000	1.3195201	18.5200000	22.5400000	7
9	8	21.2750000	1.6846958	18.1400000	23.1400000	8

Table 1. Descriptive Statistics of Weight Gain by Treatment Group

<i>Treatment</i>	<i>N</i>	<i>Mean</i>	<i>Std Dev</i>	<i>Minimum</i>	<i>Maximum</i>
1	8	22.418	2.447	18.750	26.380
2	8	21.542	1.733	19.130	23.950
3	7	20.343	2.302	17.300	23.960
4	8	19.964	0.748	19.020	21.160
5	7	20.777	2.287	16.440	23.710
6	8	21.946	1.966	18.680	25.040
7	8	19.965	1.552	17.380	21.980
8	7	19.890	1.320	18.520	22.540
9	8	21.275	1.685	18.140	23.140

Table 1. Descriptive Statistics of Weight Gain by Treatment Group

<i>Model Information</i>	
<i>Data Set</i>	WORK.WEIGHTDATA
<i>Dependent Variable</i>	wgtgain
<i>Covariance Structure</i>	Variance Components
<i>Subject Effect</i>	litter
<i>Estimation Method</i>	REML
<i>Residual Variance Method</i>	Profile
<i>Fixed Effects SE Method</i>	Kenward-Roger
<i>Degrees of Freedom Method</i>	Kenward-Roger

<i>Class Level Information</i>	
<i>Class</i>	<i>Levels</i> <i>Values</i>
<i>trt</i>	9 1 2 3 4 5 6 7 8 9
<i>litter</i>	8 1 2 3 4 5 6 7 8

<i>Dimensions</i>	
<i>Covariance Parameters</i>	2
<i>Columns in X</i>	10
<i>Columns in Z per Subject</i>	1
<i>Subjects</i>	8
<i>Max Obs per Subject</i>	9

<i>Number of Observations</i>	
<i>Number of Observations Read</i>	69
<i>Number of Observations Used</i>	69
<i>Number of Observations Not Used</i>	0

<i>Iteration History</i>			
<i>Iteration</i>	<i>Evaluations</i>	<i>-2 Res Log Like</i>	<i>Criterion</i>
0	1	262.10063124	
1	2	254.09555581	0.00000028
2	1	254.09553552	0.00000000

Convergence criteria met.

<i>Covariance Parameter Estimates</i>		
<i>Cov Parm</i>	<i>Subject</i>	<i>Estimate</i>
<i>Intercept</i>	<i>litter</i>	0.8822
<i>Residual</i>		2.5368

<i>Fit Statistics</i>	
<i>-2 Res Log Likelihood</i>	254.1
<i>AIC (Smaller is Better)</i>	258.1
<i>AICC (Smaller is Better)</i>	258.3
<i>BIC (Smaller is Better)</i>	258.3

Table 1. Descriptive Statistics of Weight Gain by Treatment Group

Solution for Fixed Effects								
Effect	trt	Estimate	Standard Error	DF	t Value	Pr > t		
Intercept		21.2750	0.6537	39.6	32.54	<.0001		
trt	1	1.1425	0.7964	52.9	1.43	0.1573		
trt	2	0.2675	0.7964	52.9	0.34	0.7383		
trt	3	-0.9310	0.8274	53.1	-1.13	0.2656		
trt	4	-1.3113	0.7964	52.9	-1.65	0.1056		
trt	5	-0.3414	0.8278	53.2	-0.41	0.6817		
trt	6	0.6712	0.7964	52.9	0.84	0.4031		
trt	7	-1.3100	0.7964	52.9	-1.64	0.1059		
trt	8	-1.2285	0.8278	53.2	-1.48	0.1437		
trt	9	0	.	.	.	.		

Type 3 Tests of Fixed Effects					
Effect	Num DF	Den DF	F Value	Pr > F	
trt	8	53.1	2.59	0.0182	

Least Squares Means						
Effect	trt	Estimate	Standard Error	DF	t Value	Pr > t
trt	1	22.4175	0.6537	39.6	34.29	<.0001
trt	2	21.5425	0.6537	39.6	32.95	<.0001
trt	3	20.3440	0.6912	43.6	29.43	<.0001
trt	4	19.9637	0.6537	39.6	30.54	<.0001
trt	5	20.9336	0.6917	43.5	30.27	<.0001
trt	6	21.9463	0.6537	39.6	33.57	<.0001
trt	7	19.9650	0.6537	39.6	30.54	<.0001
trt	8	20.0465	0.6917	43.5	28.98	<.0001
trt	9	21.2750	0.6537	39.6	32.54	<.0001

Differences of Least Squares Means									
Effect	trt	_trt	Estimate	Standard Error	DF	t Value	Pr > t	Adjustment	Adj P
trt	1	2	0.8750	0.7964	52.9	1.10	0.2769	Tukey-Kramer	0.9719
trt	1	3	2.0735	0.8274	53.1	2.51	0.0153	Tukey-Kramer	0.2526
trt	1	4	2.4537	0.7964	52.9	3.08	0.0033	Tukey-Kramer	0.0731
trt	1	5	1.4839	0.8278	53.2	1.79	0.0787	Tukey-Kramer	0.6866
trt	1	6	0.4713	0.7964	52.9	0.59	0.5565	Tukey-Kramer	0.9996
trt	1	7	2.4525	0.7964	52.9	3.08	0.0033	Tukey-Kramer	0.0733
trt	1	8	2.3710	0.8278	53.2	2.86	0.0060	Tukey-Kramer	0.1213
trt	1	9	1.1425	0.7964	52.9	1.43	0.1573	Tukey-Kramer	0.8791
trt	2	3	1.1985	0.8274	53.1	1.45	0.1534	Tukey-Kramer	0.8734
trt	2	4	1.5788	0.7964	52.9	1.98	0.0526	Tukey-Kramer	0.5619
trt	2	5	0.6089	0.8278	53.2	0.74	0.4652	Tukey-Kramer	0.9980
trt	2	6	-0.4037	0.7964	52.9	-0.51	0.6143	Tukey-Kramer	0.9999
trt	2	7	1.5775	0.7964	52.9	1.98	0.0528	Tukey-Kramer	0.5629

Table 1. Descriptive Statistics of Weight Gain by Treatment Group

<i>Differences of Least Squares Means</i>									
<i>Effect</i>	<i>trt</i>	<i>_trt</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>DF</i>	<i>t Value</i>	<i>Pr > t </i>	<i>Adjustment</i>	<i>Adj P</i>
<i>trt</i>	2	8	1.4960	0.8278	53.2	1.81	0.0764	Tukey-Kramer	0.6772
<i>trt</i>	2	9	0.2675	0.7964	52.9	0.34	0.7383	Tukey-Kramer	1.0000
<i>trt</i>	3	4	0.3803	0.8274	53.1	0.46	0.6477	Tukey-Kramer	0.9999
<i>trt</i>	3	5	-0.5896	0.8587	53.5	-0.69	0.4953	Tukey-Kramer	0.9988
<i>trt</i>	3	6	-1.6022	0.8274	53.1	-1.94	0.0581	Tukey-Kramer	0.5926
<i>trt</i>	3	7	0.3790	0.8274	53.1	0.46	0.6488	Tukey-Kramer	0.9999
<i>trt</i>	3	8	0.2976	0.8587	53.5	0.35	0.7303	Tukey-Kramer	1.0000
<i>trt</i>	3	9	-0.9310	0.8274	53.1	-1.13	0.2656	Tukey-Kramer	0.9677
<i>trt</i>	4	5	-0.9699	0.8278	53.2	-1.17	0.2466	Tukey-Kramer	0.9590
<i>trt</i>	4	6	-1.9825	0.7964	52.9	-2.49	0.0160	Tukey-Kramer	0.2604
<i>trt</i>	4	7	-0.00125	0.7964	52.9	-0.00	0.9988	Tukey-Kramer	1.0000
<i>trt</i>	4	8	-0.08271	0.8278	53.2	-0.10	0.9208	Tukey-Kramer	1.0000
<i>trt</i>	4	9	-1.3113	0.7964	52.9	-1.65	0.1056	Tukey-Kramer	0.7748
<i>trt</i>	5	6	-1.0126	0.8278	53.2	-1.22	0.2266	Tukey-Kramer	0.9476
<i>trt</i>	5	7	0.9686	0.8278	53.2	1.17	0.2472	Tukey-Kramer	0.9593
<i>trt</i>	5	8	0.8871	0.8513	52.9	1.04	0.3021	Tukey-Kramer	0.9797
<i>trt</i>	5	9	-0.3414	0.8278	53.2	-0.41	0.6817	Tukey-Kramer	1.0000
<i>trt</i>	6	7	1.9812	0.7964	52.9	2.49	0.0160	Tukey-Kramer	0.2612
<i>trt</i>	6	8	1.8998	0.8278	53.2	2.30	0.0257	Tukey-Kramer	0.3635
<i>trt</i>	6	9	0.6712	0.7964	52.9	0.84	0.4031	Tukey-Kramer	0.9949
<i>trt</i>	7	8	-0.08146	0.8278	53.2	-0.10	0.9220	Tukey-Kramer	1.0000
<i>trt</i>	7	9	-1.3100	0.7964	52.9	-1.64	0.1059	Tukey-Kramer	0.7757
<i>trt</i>	8	9	-1.2285	0.8278	53.2	-1.48	0.1437	Tukey-Kramer	0.8578

Table 2. Covariance Parameter Estimates (Random Effect of Litter)

Covariance Parameter	Subject	Estimate
Intercept	litter	0.8822
Residual		2.5368

Table 3. Fixed Effects Estimates - PROC MIXED (REML)

<i>Effect</i>	<i>Treatment</i>	<i>Estimate</i>	<i>Std Error</i>	<i>DF</i>	<i>t Value</i>	<i>p Value</i>
Intercept	_	21.2750	0.6537	39.6	32.54	0.0000
trt	1	1.1425	0.7964	52.9	1.43	0.1573
trt	2	0.2675	0.7964	52.9	0.34	0.7383
trt	3	-0.9310	0.8274	53.1	-1.13	0.2656
trt	4	-1.3113	0.7964	52.9	-1.65	0.1056
trt	5	-0.3414	0.8278	53.2	-0.41	0.6817
trt	6	0.6712	0.7964	52.9	0.84	0.4031
trt	7	-1.3100	0.7964	52.9	-1.64	0.1059
trt	8	-1.2285	0.8278	53.2	-1.48	0.1437
trt	9	0.0000	.	.	.	.

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

Effect	Num		Den		F Value	p Value
	DF	DF	DF	DF		
trt	8	53.1	2.589	0.0182		

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

Class Level Information										
Class	Levels	Values								
<i>trt</i>	9	1 2 3 4 5 6 7 8 9								
<i>litter</i>	8	1 2 3 4 5 6 7 8								

<i>Number of Observations Read</i>	69
<i>Number of Observations Used</i>	69

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect**Dependent Variable: wgtgain**

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	15	124.9518071	8.3301205	3.29	0.0007
Error	53	134.3274915	2.5344810		
Corrected Total	68	259.2792986			

R-Square	Coeff Var	Root MSE	wgtgain Mean
0.481920	7.607491	1.592005	20.92681

Source	DF	Type I SS	Mean Square	F Value	Pr > F
trt	8	54.98293784	6.87286723	2.71	0.0140
litter	7	69.96886926	9.99555275	3.94	0.0016

Source	DF	Type III SS	Mean Square	F Value	Pr > F
trt	8	51.75790299	6.46973787	2.55	0.0197
litter	7	69.96886926	9.99555275	3.94	0.0016

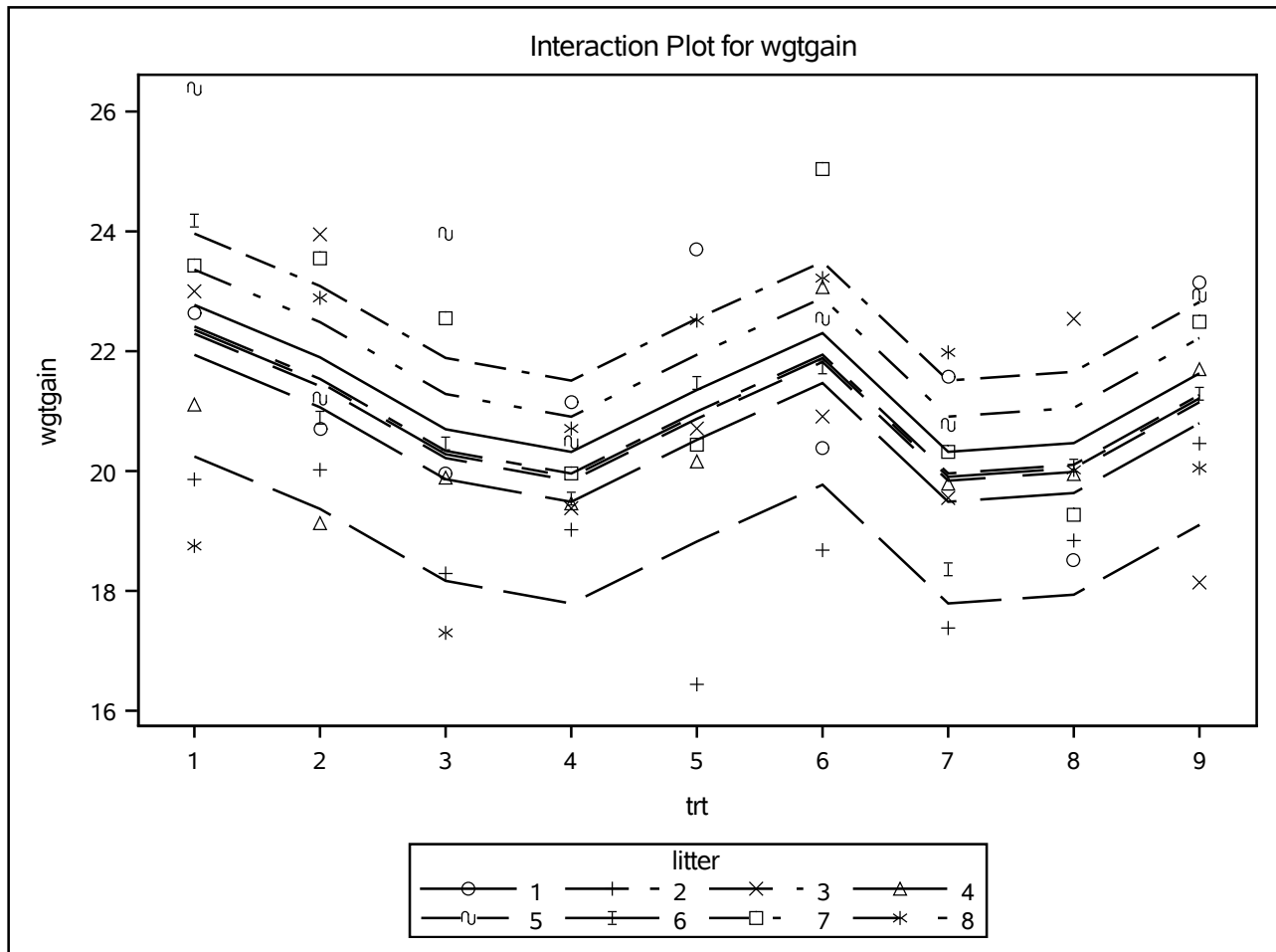


Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect**Variable: resid**

Moments			
N	69	Sum Weights	69
Mean	0	Sum Observations	0
Std Deviation	1.40549076	Variance	1.97540429
Skewness	-0.2389607	Kurtosis	-0.289214
Uncorrected SS	134.327491	Corrected SS	134.327491
Coeff Variation	.	Std Error Mean	0.1692012

Basic Statistical Measures			
Location		Variability	
Mean	0.000000	Std Deviation	1.40549
Median	0.034878	Variance	1.97540
Mode	.	Range	5.97591
		Interquartile Range	2.01125

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > t	1.0000
Sign	M	1.5	Pr >= M	0.8099
Signed Rank	S	22.5	Pr >= S	0.8941

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.980992	Pr < W	0.3761
Kolmogorov-Smirnov	D	0.05578	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.030027	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.248981	Pr > A-Sq	>0.2500

Quantiles (Definition 5)	
Level	Quantile
100% Max	2.4340638
99%	2.4340638
95%	2.3566243
90%	2.0724289
75% Q3	1.0642615
50% Median	0.0348783
25% Q1	-0.9469885
10%	-1.9217107
5%	-2.3833757
1%	-3.5418496
0% Min	-3.5418496

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

Variable: resid			
Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
-3.54185	61	2.15051	57
-3.13016	26	2.35662	5
-2.91652	63	2.41234	20
-2.38338	14	2.41710	36
-1.94623	8	2.43406	25

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

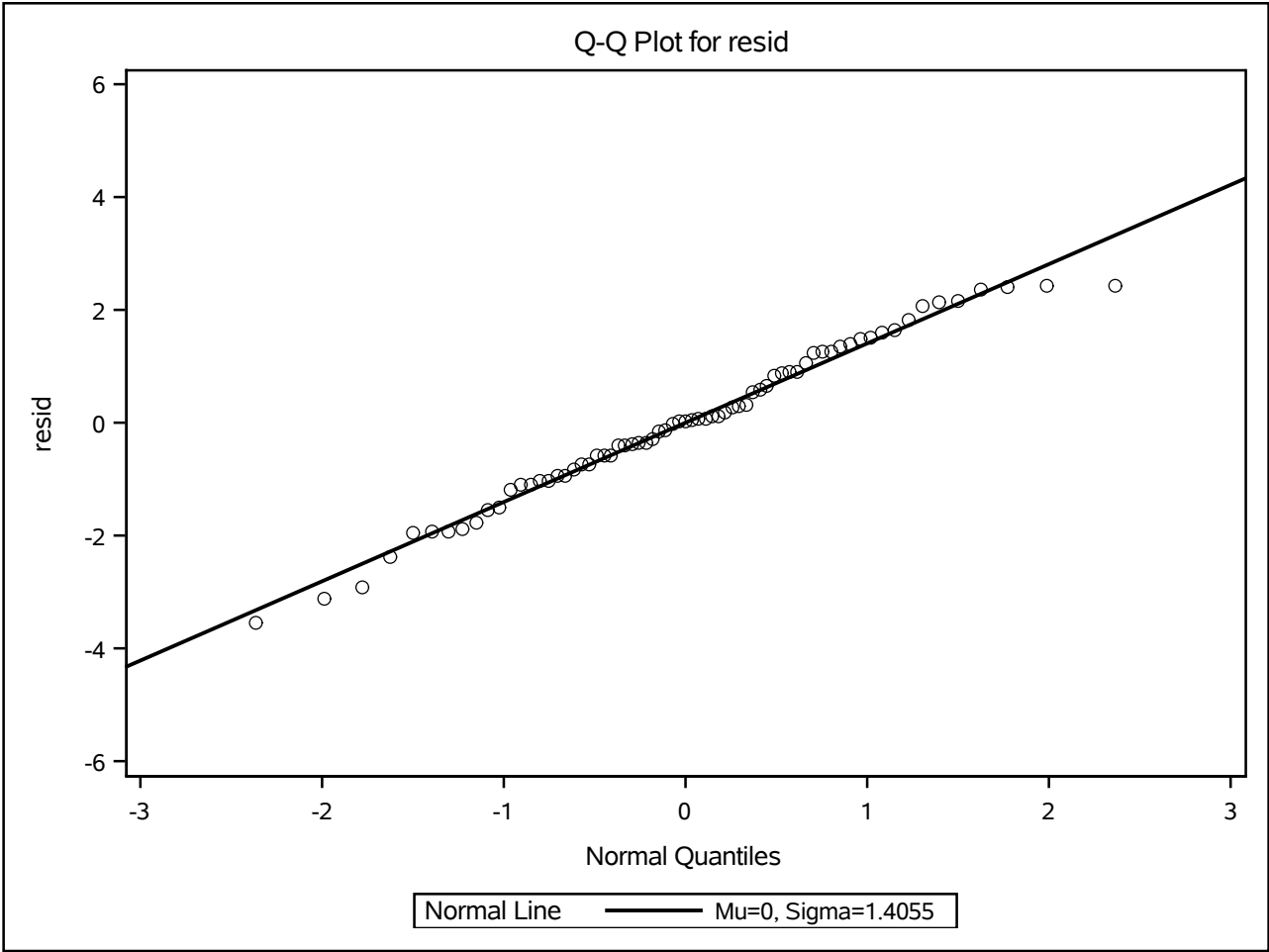


Table 5. Tests for Normality of Residuals (Shapiro-Wilk)

<i>Test</i>	<i>Statistic</i>	<i>p Value</i>
Shapiro-Wilk	0.980992	0.3761
Kolmogorov-Smirnov	0.05578	0.1500
Cramer-von Mises	0.030027	0.2500
Anderson-Darling	0.248981	0.2500

Table 5. Tests for Normality of Residuals (Shapiro-Wilk)

Class Level Information									
Class	Levels	Values							
trt	9	1 2 3 4 5 6 7 8 9							
Number of Observations Read		69							
Number of Observations Used		69							

Table 5. Tests for Normality of Residuals (Shapiro-Wilk)**Dependent Variable: wgtgain**

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	54.9829378	6.8728672	2.02	0.0593
Error	60	204.2963607	3.4049393		
Corrected Total	68	259.2792986			

R-Square	Coeff Var	Root MSE	wgtgain Mean
0.212061	8.817625	1.845248	20.92681

Source	DF	Type I SS	Mean Square	F Value	Pr > F
trt	8	54.98293784	6.87286723	2.02	0.0593

Source	DF	Type III SS	Mean Square	F Value	Pr > F
trt	8	54.98293784	6.87286723	2.02	0.0593

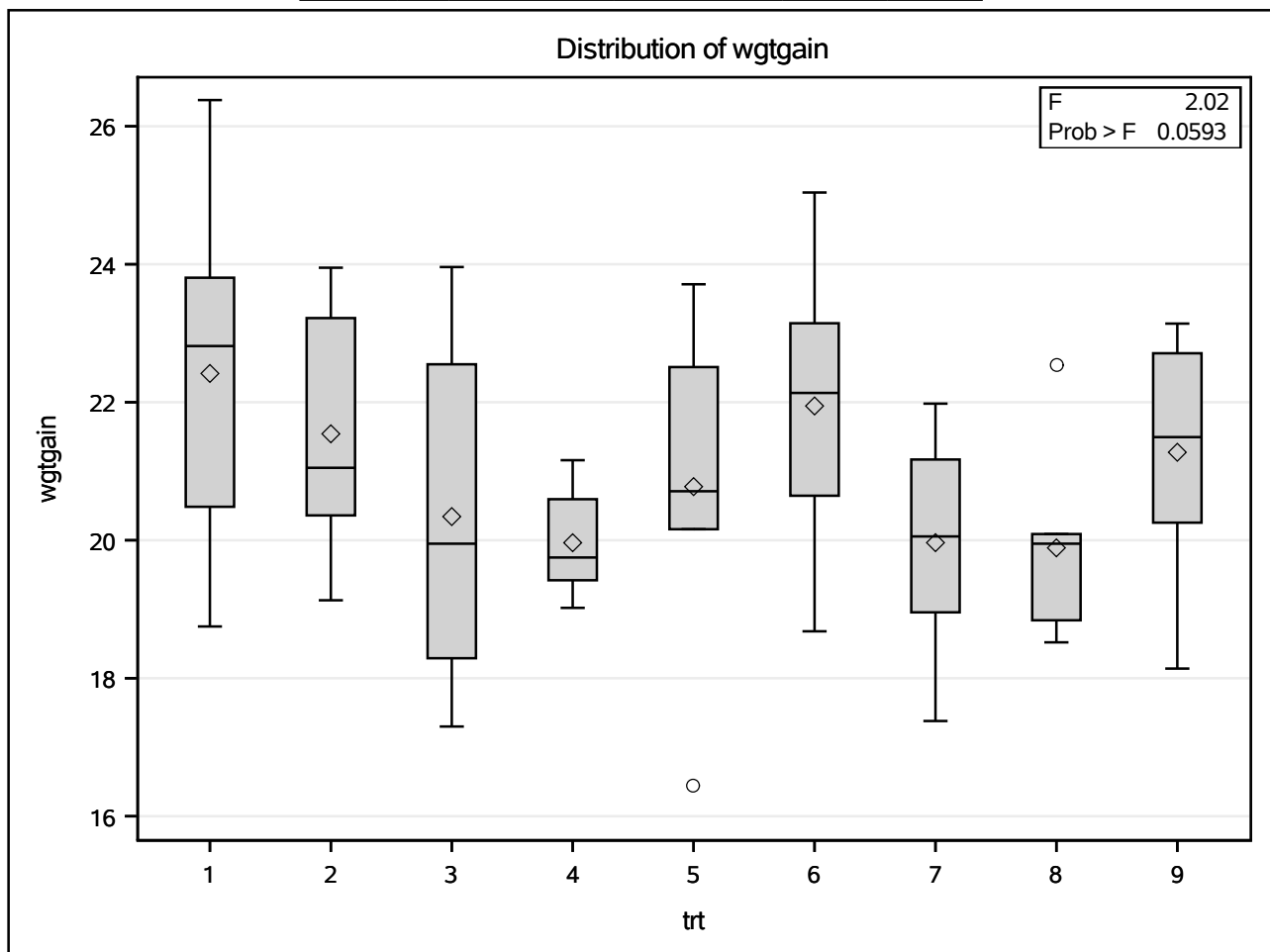
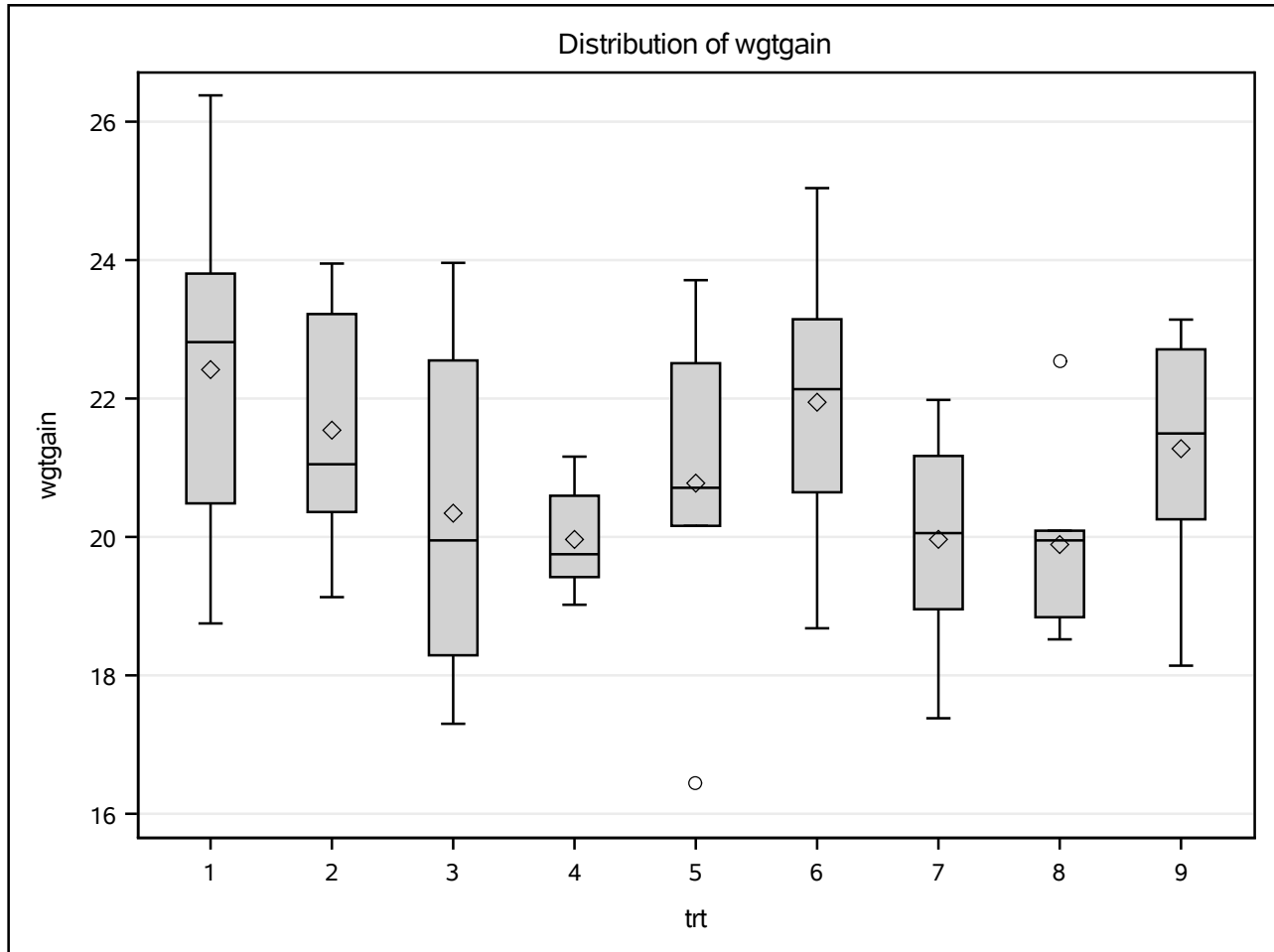


Table 5. Tests for Normality of Residuals (Shapiro-Wilk)

<i>Levene's Test for Homogeneity of wgtgain Variance</i>					
<i>ANOVA of Squared Deviations from Group Means</i>					
<i>Source</i>	<i>DF</i>	<i>Sum of Squares</i>	<i>Mean Square</i>	<i>F Value</i>	<i>Pr > F</i>
<i>trt</i>	8	149.2	18.6501	1.09	0.3845
<i>Error</i>	60	1029.2	17.1540		

Table 5. Tests for Normality of Residuals (Shapiro-Wilk)

wgtgain			
Level of trt	N	Mean	Std Dev
1	8	22.4175000	2.44710762
2	8	21.5425000	1.73312888
3	7	20.3428571	2.30222873
4	8	19.9637500	0.74825392
5	7	20.7771429	2.28744191
6	8	21.9462500	1.96582180
7	8	19.9650000	1.55248188
8	7	19.8900000	1.31952011
9	8	21.2750000	1.68469582

Table 6. Levene's Test for Homogeneity of Variance

Source	DF	Sum of Squares	Mean Square	F Value	p Value
trt	8	149.2	18.6501	1.087	0.3845
Error	60	1029.2	17.1540	—	—

Table 6. Levene's Test for Homogeneity of Variance

Class Level Information										
Class	Levels	Values								
trt	9	1 2 3 4 5 6 7 8 9								
litter	8	1 2 3 4 5 6 7 8								

Number of Observations Read	69
Number of Observations Used	69

Table 6. Levene's Test for Homogeneity of Variance**Dependent Variable: wgtgain**

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	15	124.9518071	8.3301205	3.29	0.0007
Error	53	134.3274915	2.5344810		
Corrected Total	68	259.2792986			

R-Square	Coeff Var	Root MSE	wgtgain Mean
0.481920	7.607491	1.592005	20.92681

Source	DF	Type I SS	Mean Square	F Value	Pr > F
trt	8	54.98293784	6.87286723	2.71	0.0140
litter	7	69.96886926	9.99555275	3.94	0.0016

Source	DF	Type III SS	Mean Square	F Value	Pr > F
trt	8	51.75790299	6.46973787	2.55	0.0197
litter	7	69.96886926	9.99555275	3.94	0.0016

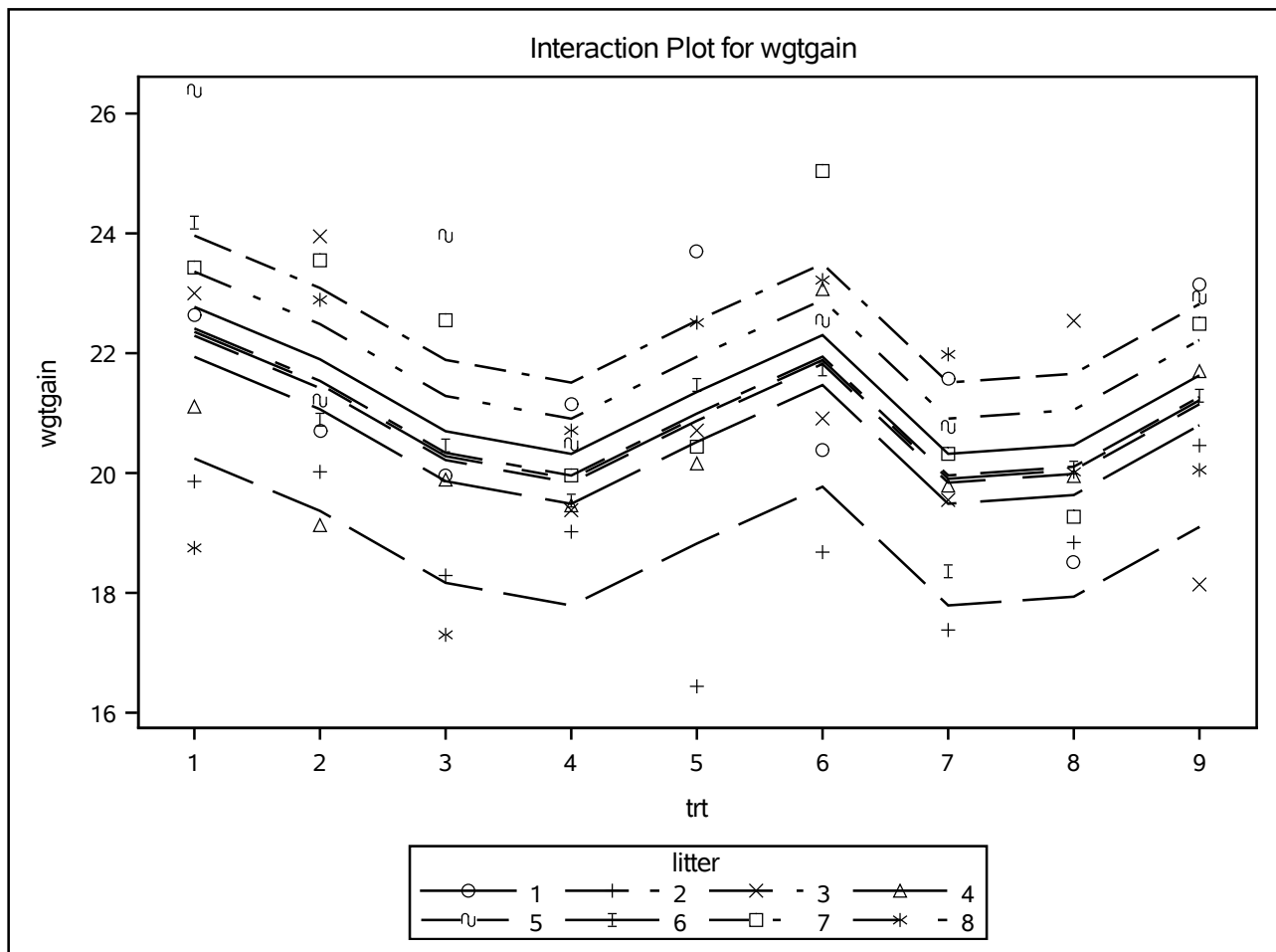


Table 7. Comparison: Traditional ANOVA (Litter as Fixed Effect)

<i>Source</i>	<i>DF</i>	<i>Sum of Squares</i>	<i>Mean Square</i>	<i>F Value</i>	<i>p Value</i>
trt	8	54.98293784	6.87286723	2.712	0.0140
litter	7	69.96886926	9.99555275	3.944	0.0016
trt	8	51.75790299	6.46973787	2.553	0.0197
litter	7	69.96886926	9.99555275	3.944	0.0016